

2. (a) $p_1 V_1 = p_2 V_2$ (Or $p \propto \frac{1}{V}$) 1M
- $$p_1 \left(\frac{4}{3} \pi \times (0.8)^3 \right) = (1.01 \times 10^5) \left(\frac{4}{3} \pi \times (1.0)^3 \right)$$
- $$p_1 = 1.97 \times 10^5 \text{ Pa}$$
- 1A 2
- (b) Volume increases as bubble rises but the speed / k.e. of gas molecules remains unchanged 1A
 therefore frequency of collision of molecules on bubble's inner surface decreases, gas pressure decreases. 1A 2

- | | | |
|--|-----------------------|--|
| 2. (a) $210 \text{ atm} \times (1.0 \times 10^4 \text{ cm}^3) = 2.0 \text{ atm} \times V$
$V = 1.05 \times 10^6 \text{ cm}^3$
Volume available = $1.05 \times 10^6 - 1.0 \times 10^4$
$= 1.04 \times 10^6 \text{ (cm}^3\text{)}$ | 1M

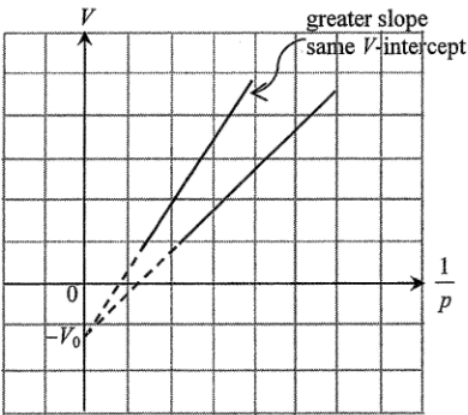
1A

2 | Accept ans. without considering residual volume, i.e. $1.05 \times 10^6 \text{ (cm}^3\text{)}$ |
| (b) (i) $V_0 = 1.04 \times 10^6 \text{ cm}^3 \div 60$
$= 17333 \approx 17300 \text{ (cm}^3\text{) (per minute)}$ | 1M/1A

1 | $V_0 = 17500 \text{ (cm}^3\text{) if residual volume not considered.}$ |
| (ii) V' : total volume of air at this depth/in this situation
$\frac{P_1 V_1}{T_1} = \frac{P_2 V'}{T_2}$ $\frac{210 \times (1.0 \times 10^4)}{273 + 24} = \frac{4.5 \times V'}{273 + 20}$ $V' = 4.60 \times 10^5 \text{ cm}^3$ Volume available = $4.60 \times 10^5 - 1.0 \times 10^4$
$= 4.50 \times 10^5 \text{ (cm}^3\text{)}$
Length of time :
$= \frac{4.50 \times 10^5}{17333}$
$= 26.0 \text{ (min.)}$ | 1M

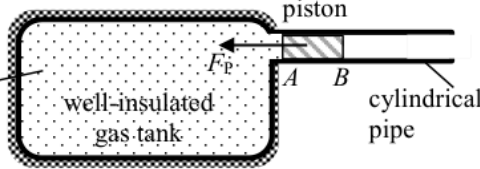
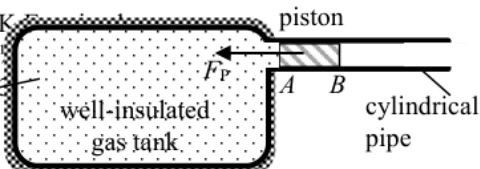
1A

3 | $V' = 4.60 \times 10^5 \text{ cm}^3$ and length of time = 26.3 min. if residual volume not considered. |

2. (a) $pV = nRT$ $(1.0 \times 10^5)(6.0 \times 10^{-5}) = n(8.31)(273 + 25)$ $n = 2.422891 \times 10^{-3} \text{ moles} \approx 2.42 \times 10^{-3} \text{ moles}$ No. of molecules $= nN_A$ $= n \times 6.02 \times 10^{23}$ $= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$ <i>Alternative method:</i> $pV = nRT = \left(\frac{N}{N_A}\right)RT \Rightarrow N = \left(\frac{pV}{RT}\right)N_A$ $N = \frac{(1.0 \times 10^5)(6.0 \times 10^{-5})}{(8.31)(273 + 25)} \times (6.02 \times 10^{23})$ $= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$	1M 1M 1A 1M 1M 1A	
(b) (i) - The piston should be pushed or pulled slowly / lightly. - Avoid taking readings immediately after moving the piston. - The syringe should not be grasped by hand too long (when the piston is pushed in or pulled out).	3 1A 1A 1A Any ONE	
(ii) V_0 - volume of gas trapped in the rubber tubing / space connecting the pressure sensor and syringe.	1 1A	
(iii) 	1 1A 1A 2	NO mark for a curve

Or $(2p)(2V) = 0.8RT$ $pV = 0.2RT$ $p(3V) = nRT$ $n = 3 \times 0.2 = 0.6 \text{ (mol)}$	1M	
	1A	
		2
(b) (i) $n = n_A + n_B$ $p'(3V + 2V) = p(3V) + (2p)(2V)$ $p' = 1.4p$	1M	
	1A	
Or $p'(3V + 2V) = (0.6 + 0.8)RT$ $p'(5V) = 1.4RT$ $p' = 1.4p$	1M	
	1A	
		2
(ii) When the tap is open, number of gas molecules in vessel A increases due to the (net) flow of molecules from B to A, the frequency of collision of gas molecules with the vessel's wall increases, pressure increases.	1A	
	1A	
		2

2. (a) $pV = nRT$ $(100 \times 10^3)(0.52) = n(8.31)(273 + 15)$ $n = 21.727504 \text{ (mol)} \approx 21.7 \text{ (mol)}$	1M	
	1A	Accept: (21.0 ~ 22.0) mol
		2
(b) (i) Since $pV = nRT \Rightarrow V = \frac{nRT}{p}$ volume V of the balloon depends on both T and p , the (fractional) decrease in pressure p (with height) is greater / faster than the (fractional) decrease in temperature T .	1A	Accept: volume V decreases as temperature T decreases would be true only if pressure p is constant, however, p decreases with height.
	1A	
		2
(ii) (1) $\frac{pV}{T} = \text{constant}$ $\frac{(100)(0.52)}{(273 + 15)} = \frac{p(8)}{216}$ $p = 4.875 \text{ kPa or } 4875 \text{ Pa}$	1M	
	1A	
		2
(2) $p = p_0 e^{-kx}$ $4.875 = 100 e^{-0.138x}$ $x = 21.89166726 \text{ (km)} \approx 21.9 \text{ (km)}$	1M	
	1A	Accept: (21.7 ~ 22.0) km
		2

2. (a) (i)  high-pressure steam (237 °C, 3.10×10^6 Pa) well-insulated gas tank piston F_P A B cylindrical pipe	1A	F_P correctly marked F_P should be in contact with the piston. Accept: no label of the force
(ii) $F_P = (3.10 \times 10^6 - 1.0 \times 10^5) \times 0.67$ $= 2010000 \text{ N} = 2.01 \times 10^6 \text{ N}$	1 1M 1A 2	Accept: $F_P = (3.10 \times 10^6) \times 0.67$ Accept $2.00 \times 10^6 \text{ N}$ to $2.01 \times 10^6 \text{ N}$
(iii) $pV = nRT \Rightarrow V = \frac{nRT}{p}$ $V = \frac{(570/0.018)(8.31)(237 + 273)}{3.10 \times 10^6}$ $= 43.292419 \text{ m}^3 \approx 43.3 \text{ m}^3$	1M+1M 1A 3	1st 1M find n (570/0.018) 2nd 1M $pV=nRT$, accept $T=237$ Accept 43.0 m^3 to 43.3 m^3
(b) (i) Work done = K  high-pressure steam (237 °C, 3.10×10^6 Pa) well-insulated gas tank piston F_P A B cylindrical pipe	1M 1A 2	Accept 37.9 MJ to 38.0 MJ Accept: $a = 54/1.5 = 36 \text{ (m s}^{-2}\text{)}$ $F = (2.6 \times 10^4) \times 36$ $= 936000 \text{ (N)}$ $s = 54^2 / (2 \times 36) = 40.5 \text{ (m)}$ [or $v = at/2 = 36(1.5)/2$ $= 27 \text{ (m s}^{-1}\text{)}$ $W = Fs = 936000 \times 340.5$ [or $W = Fvt = 936000 \times 27 \times 1.5$ $= 37908000 \text{ (J)}$ [1M+1A] If $s = vt$ [0M 0A]
(ii) Average acceleration $a = \frac{v-u}{t} = \frac{54-0}{1.5}$ $= 36 \text{ m s}^{-2}$	1M 1A 2	
(iii) Acceleration is decreasing (i.e. maximum at first). (According to kinetic theory,) once the steam expands, i.e. volume increases, its pressure decreases, thus the (pressure) force acting on the piston at A decreases, hence a smaller acceleration.	1A 1A 1A 3	1st 1A for acceleration decreasing 2nd 1A for <u>volume increases/</u> <u>length increases</u> $\Rightarrow$ <u>pressure decreases</u> Alternative method: The acceleration of the jet fighter is decreasing, 1A As the <u>air resistance</u> (f) acts on the jet fighter <u>increases with the speed.</u> 1A The net force ($ma = F - f$) is decreasing. 1A

2.	(a)	(i)	(I)	$pV = nRT$ $(1.0 \times 10^5) (6.0 \times 10^{-4}) = n (8.31) (300)$ $n = 0.0240674$ number of molecules $N = nN_A$ $= (0.0240674) (6.02 \times 10^{23})$ $= 1.448857 \times 10^{22} \approx 1.45 \times 10^{22}$	1M	
					1A	
			(II)	average kinetic energy of gas molecules		
				$E_K = \frac{3}{2} \left(\frac{R}{N_A} \right) T$ $= \frac{3}{2} \left(\frac{8.31}{6.02 \times 10^{23}} \right) (300)$ $= 6.211794 \times 10^{-21} \text{ J} \approx 6.21 \times 10^{-21} \text{ J}$	1M	
					1A	
						4
	(ii)	(I)		N and E_K remain unchanged.	1A+1A	
						2
			(II)	$E_K = \frac{1}{2} m c_{r.m.s.}^2$ $\frac{1}{2} m (600)^2 = \frac{1}{2} \left(\frac{1}{5} m \right) c_{r.m.s.}^2$ $\Rightarrow c_{r.m.s.} = \sqrt{5} \times 600$ $= 1341.6408 \text{ m s}^{-1} \approx 1340 \text{ m s}^{-1}$	1M	
					1A	
						2
	(b)	As gas C diffuses into the upper jar, its molecules collide with the air molecules and thus describe a zig-zag path / not along a straight path.			1A	
					1A	
						2

2. (a) $m_0 = \text{density} \times \text{volume}$ $= 1.20 \times (1.5 \times 10^{-3})$ $= 1.8 \times 10^{-3} \text{ kg or } 1.8 \text{ g}$	1M 1A	
	2	
(b) (i) $p_1 V = n_1 R T$ and $p_2 V = n_2 R T$ As $m_0 \rightarrow 4m_0$, $n_2 = 4n_1$ $\therefore p_2 = \frac{n_2 R T}{V} = \frac{4n_1 R T}{V} = 4p_1 = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A	
Or $pV = nRT$ $p \propto n$ (as V , R and T are constant) Since $m_0 \rightarrow 4m_0$, thus $n \rightarrow 4n$, p becomes 4 times the original value (= 1.0 atm), $p = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A	
	2	
(ii) More air molecules hit unit area of surface of wall or bottle / water surface (as $n \rightarrow 4n$ moles of molecules trapped in the same volume), The surface of wall or bottle / water surface is hit more frequently OR rate of change of total momentum, i.e. force, per unit area of wall or bottle / water surface increases, pressure increases.	1A 1A	
	2	
(iii) $\Delta F = \text{increase in pressure} \times \text{area}$ $= [(4 - 1) \times 10^5] \times 0.014$ $= 4200 \text{ N}$	1M 1A	error-carried-forward (e.c.f.) from (b)(i)
	2	
(c) (i) The pressure force acting on the water results in its change of momentum (a water jet). Magnitude of the force = rate of change of momentum. According to Newton's third law of motion, an equal but opposite force / reaction / thrust acts on the rocket. (According to Newton's second law of motion,) the reaction force causes the rocket to accelerate / move upwards.	1A 1A	
	2	
(ii) kinetic energy of (the jet of) water / work done on (the jet of) water	1A	
	1	

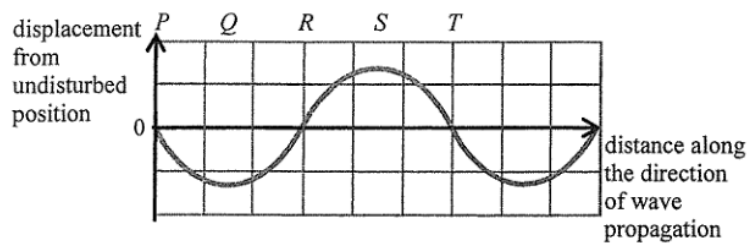
$$= 3 \left(\frac{8.31}{\text{mol}} \right) \left(\frac{1}{2} \right) (400.247 \text{ K})$$

$$= 400.247 \text{ (K)} \approx 400 \text{ (K)}$$

- (b) As temperature remains at T_0 ,
 E_K remains unchanged / $c_{r.m.s.}$ depends on temperature,
 so root-mean-square speed $c_{r.m.s.}$ remains unchanged.

1A
2
1A
1A
2

5. (a)



- (b) (i) $P / R / T$
- (ii) R

2A
2
1A
1
1A
1

(i) (a) $pV = nRT$ $1.01 \times 10^5 \times V = 1 \times 8.31 \times 273$ $V = 2.2462 \times 10^{-2} \text{ m}^3 \approx 2.25 \times 10^{-2} \text{ m}^3$ (ii) Number of molecules in 1 m^3 $= \frac{N_A}{\text{volume of one mole of molecule}}$ $= \frac{6.02 \times 10^{23}}{2.246 \times 10^{-2}}$ $= 2.680120 \times 10^{25} \approx 2.68 \times 10^{25}$ (b) $pV = \left(\frac{N}{N_A} \right) RT$ $2.75 \times 10^{-10} \times 1 = \left(\frac{1.00 \times 10^{11}}{6.02 \times 10^{23}} \right) \times 8.31 \times T$ $T = 199.2178 \text{ (K)} \approx 199 \text{ (K)}$	1M 1A 2	Accept: $0.0220 \sim 0.0225 \text{ m}^3$
	1M 1A 2	e.c.f. from (a)(i) Or $pV = nRT$ $1.01 \times 10^5 \times 1 = n \times 8.31 \times 273$ $n = 44.5$ $N = 44.5 \times 6.02 \times 10^{23}$ $= 2.68 \times 10^{25}$
	1M 1A 2	Accept: $199 \sim 200 \text{ K}$